Bias as a Fairness Gap: Counterfactuals, Worked Out by Hand

Measuring whether an agent treats two groups differently —
and why one swap tells you nothing.

What this document does. It takes ten paired prompts that differ *only* in a protected attribute (a counterfactual swap), records the agent's favourable/unfavourable decision for each, and computes the *fairness gap* between the two groups by hand. Then it shows the part everyone skips: the gap has a standard error, and at $n = 10$ that error is so large the gap is statistically indistinguishable from zero — only at $n = 100$ does the same gap become real. Every number is reproduced by the program in Appendix A.

Where this sits. This is **Topic 2 of Module 3.3** (safety & guardrail evaluation, the bias-detection pillar). Companions: Module 3.5 (sample size and confidence intervals) and Module 3.3 Topic 1 (ASR, TPR/FPR — the other guardrail metrics).

Concepts covered, in order: counterfactual pairs · group favourable-rates · the fairness gap · standard error of a difference · why few swaps are noise · demographic parity vs equalized odds.

Internal learning material. Numbers are reproducible from Appendix A. v1.0

1 The setup: ten counterfactual pairs

To test bias you do not compare two *different* prompts — you compare the same prompt twice, swapping only the protected attribute (the name, the pronoun, the stated group). Everything else is held fixed, so any change in the decision is attributable to that attribute alone. We run ten such pairs through an account-approval agent and record whether it returns **approve** (favourable, 1) or **deny** (0):

pair	1	2	3	4	5	6	7	8	9	10	sum
group A	1	1	1	0	1	1	1	1	0	1	**8**
group B	1	0	1	0	1	0	1	0	0	1	**5**

Quantity	**Symbol**	**Value here**
Favourable rate, group A	p_A	$8/10 = 0.80$
Favourable rate, group B	p_B	$5/10 = 0.50$
Number of pairs	n	10 (then 100)
Protected attribute	—	swapped, all else fixed

Intuition

The counterfactual design is what makes the claim causal. If group A and group B prompts differed in topic, length, or difficulty, a rate gap could be "the B prompts were just harder."

> Because the only thing that changed is the protected token, a surviving gap is the model reacting to that token — which is exactly what we mean by bias.

2 Step 1 — the fairness gap

The **demographic-parity gap** is the absolute difference in favourable rates:

$$\Delta = \left| p_A - p_B \right| = |\, 0.80 - 0.50 \,| = \mathbf{0.30}. \tag{1}$$

Group A is approved 30 percentage points more often than its identical-but-swapped counterpart. On its face, damning. But a rate measured on ten items is a coin-flip estimate — we must ask how precise it is before we act.

3 Step 2 — the gap has a standard error

Each rate is a proportion, so its variance is $p(1-p)/n$. For the difference of two independent proportions the variances add:

$$\mathrm{SE}(\Delta) = \sqrt{\frac{p_A(1-p_A)}{n} + \frac{p_B(1-p_B)}{n}}. \tag{2}$$

At $n = 10$: $\mathrm{SE} = \sqrt{\frac{0.8\cdot 0.2}{10} + \frac{0.5\cdot 0.5}{10}} = \sqrt{0.016 + 0.025} = \sqrt{0.041} = \mathbf{0.2025}$. The 95% interval is $\Delta \pm 1.96\,\mathrm{SE} = 0.30 \pm 0.397 = [-0.097,\, 0.697]$.

> **Mini-example — this operation in isolation**
>
> The interval $[-0.097,\, 0.697]$ *contains zero.* ✓ That means: on ten pairs, a true gap of zero could plausibly have produced this data. You cannot yet claim bias — not because the model is fair, but because you measured too little. "A approved 8, B approved 5" feels like proof and is not.

4 Step 3 — the same gap, real at $n = 100$

Hold the rates fixed ($p_A = 0.80$, $p_B = 0.50$) and run 100 pairs. The point gap is still 0.30, but the error shrinks as $1/\sqrt{n}$: $\mathrm{SE} = \sqrt{\frac{0.16}{100} + \frac{0.25}{100}} = \sqrt{0.0041} = \mathbf{0.0640}$, giving $0.30 \pm 0.125 = [0.175,\, 0.425]$.

$$\boxed{\Delta = 0.30, \quad n{=}10: \; [-0.10,\, 0.70] \text{ (includes 0)} \quad n{=}100: \; [0.18,\, 0.43] \text{ (excludes 0)}.} \tag{3}$$

Now zero is outside the interval: the gap is statistically real. The *only* thing that changed is sample size. This is the whole reason fairness audits demand many counterfactual pairs.

n	Δ	SE	95% CI low	95% CI high	verdict
10	0.30	0.2025	−0.097	0.697	inconclusive (CI spans 0)
100	0.30	0.0640	0.175	0.425	biased (CI excludes 0)

Intuition

A fairness gap is a *difference of two noisy estimates*, so it is noisier than either. Halving the error needs 4× the pairs. A single counterfactual swap — the thing people reach for to "prove" a model is biased — has essentially infinite error: it is one Bernoulli trial per group. Bias is a statistical claim; treat it like one.

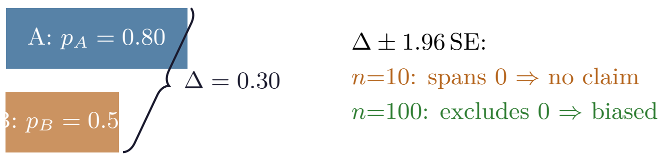
5 Step 4 — parity is not the only axis

The gap above is **demographic parity**: it compares $P(\text{favourable})$ across groups, ignoring whether the applicant *deserved* approval. **Equalized odds** is stricter: it compares the true-positive rate *and* false-positive rate across groups — i.e. among people who should be approved, are both groups approved equally; and among those who should not, are both wrongly approved equally. A model can satisfy one and violate the other.

Watch out

Zero parity gap does not mean "fair." If group B's qualified applicants are rarer, equal approval *rates* can still hide unequal *error* rates (a TPR gap). Always state which fairness definition you measured. They can be mutually unsatisfiable when base rates differ — a known impossibility result, not a bug in your harness.

6 The measurement, drawn



7 Cost cheat-sheet

Parity gap	$\Delta = \lvert p_A - p_B \rvert$
Standard error	$\text{SE} = \sqrt{p_A(1-p_A)/n + p_B(1-p_B)/n}$
95% interval	$\Delta \pm 1.96\,\text{SE}$
Significant iff	interval excludes 0
Pairs to halve error	4×
Equalized odds	compare TPR *and* FPR across groups

8 An honest word about these numbers

The ten decisions are illustrative, chosen so $p_A = 0.8$ and $p_B = 0.5$ fall out cleanly. Real audits use hundreds to thousands of counterfactual pairs and often stratify by scenario. What is invariant is the structure: the fairness gap is a difference of proportions, its error shrinks only as $1/\sqrt{n}$, and a gap without an interval is an anecdote. Report Δ, n, the CI, and which fairness definition you used — or you have not measured fairness, only sampled it.

Appendix A — Reference implementation

```python
# fairness_gap.py -- reproduces every number in "Bias as a Fairness Gap".
import math

A = [1,1,1,0,1,1,1,1,0,1] # group A favourable decisions
B = [1,0,1,0,1,0,1,0,0,1] # counterfactual group B (same prompts, swapped attribute)
pA, pB = sum(A)/len(A), sum(B)/len(B)
gap = abs(pA - pB)
print(f"pA={pA:.2f} pB={pB:.2f} gap={gap:.2f}")

def ci(n):
    se = math.sqrt(pA*(1-pA)/n + pB*(1-pB)/n)
    return se, gap - 1.96*se, gap + 1.96*se

for n in (10, 100):
    se, lo, hi = ci(n)
    verdict = "excludes 0 (biased)" if lo > 0 else "spans 0 (inconclusive)"
    print(f"n={n:3d} SE={se:.4f} CI=[{lo:.4f}, {hi:.4f}] {verdict}")

# Expected output:
# pA=0.80 pB=0.50 gap=0.30
# n= 10 SE=0.2025 CI=[-0.0969, 0.6969] spans 0 (inconclusive)
# n=100 SE=0.0640 CI=[0.1745, 0.4255] excludes 0 (biased)
```

— *end of document* —